FinInteract: The Single-Gold Illusion and the Elicitation Bottleneck in Ambiguous Question Answering

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ABSTRACT

Question-answering benchmarks grade each query against a single gold answer. When a query is genuinely ambiguous, this convention silently rewards a model for producing the *most common* reading rather than for recognizing that the query is under-specified—so a model that blindly guesses the default and one that correctly identifies the ambiguity receive the same score. We call this the *single-gold illusion.* The mechanism is domain-general—single-gold grading credits the default reading whatever the domain—so we study it where it can be measured most cleanly: finance, where the same query is routinely ambiguous (*"What was Meta Platforms' operating income?"* is \$46.75B consolidated or \$62.87B for the Family of Apps segment, a 34% gap) yet each reading's ground truth is exact and programmatically verifiable. We release FinInteract, a bilingual (English/Chinese) benchmark of 173 instances that pairs each question with a *default* interpretation (the reading a non-expert assumes) and an *intended* interpretation fixed by a disambiguating context, across a five-axis taxonomy of ambiguity. Regrading the *same* model outputs against the intended rather than the default interpretation cuts apparent accuracy by up to 3×. The failure is elicitation, not knowledge: given the resolved interpretation models answer 93–95% correctly, yet in free interaction the best model resolves only 20%, and *adding* clarification turns lowers accuracy below plain search for every model but the strongest. Finally, the required skills *dissociate*, and their trainability follows a *salience-graded data-need*: axis-guided self-training installs the salient entity axis from a single batch, the mid-salience metric axis only after accumulating several, and the least-salient axis (recognition basis) resists scale and is reached only unreliably. A per-axis skill metric turns "the model was wrong" into "the model failed to elicit axis X"—a diagnostic single-gold grading cannot provide. A replication on the general-domain AmbigQA benchmark confirms the illusion is not finance-specific: across three models, single-gold grading overstates competence over 2× there too (2.1–2.3×).

CCS CONCEPTS

• **Information systems** → **Question answering**; • **Computing methodologies** → *Natural language processing*.

KEYWORDS

evaluation methodology, measurement validity, ambiguity resolution, interactive agents, clarification, single-gold illusion, financial question answering, large language models

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1 INTRODUCTION

Question-answering benchmarks grade each query against a single gold answer. This convention is invisible when queries are fully specified—but real queries often are not, and when a query is genuinely ambiguous, single-gold grading silently rewards a model for producing the *most common* reading rather than for recognizing that the query is under-specified at all. A model that blindly guesses the default interpretation and a model that correctly identifies the ambiguity receive the *same* score. The benchmark therefore cannot see the capability that matters most in deployment: knowing when to ask.

We call this the *single-gold illusion*, and we study it where it can be measured most cleanly. Financial question answering is an ideal instrument: the same query is routinely ambiguous—*"What was Meta Platforms' operating income?"* (Figure 1) is \$46.75B as the consolidated GAAP figure for FY2023 or \$62.87B for the Family of Apps segment, a 34% difference—yet the ground truth for each reading is exact and programmatically verifiable against SEC filings. A skilled analyst would not guess; they would ask a clarifying question. Existing benchmarks reward the model that guesses correctly and cannot distinguish it from one that recognizes the ambiguity—the same blind spot, we will argue, that any single-gold benchmark inherits on ambiguous inputs, in finance or out.

Limitations of Existing Benchmarks. FinanceBench [8] provides 150 expert-curated questions over SEC filings, each with a unique correct answer verified against the source document; queries are pre-disambiguated by construction. FinQA [5] and DocFinQA [16] target numerical reasoning over earnings reports but give the model the exact evidence table alongside the question, bypassing retrieval and interaction entirely. BizBench [9] and FinBen [19] aggregate multiple financial tasks but treat QA as closed-book or retrieval-augmented reading comprehension. None of these benchmarks ask whether a model *recognizes* that a question is ambiguous, nor whether it can *resolve* that ambiguity through strategic interaction with a user.

The Ambiguous-QA Gap in Finance. The ambiguous-QA literature has matured in the general domain: AmbigQA [14] introduced

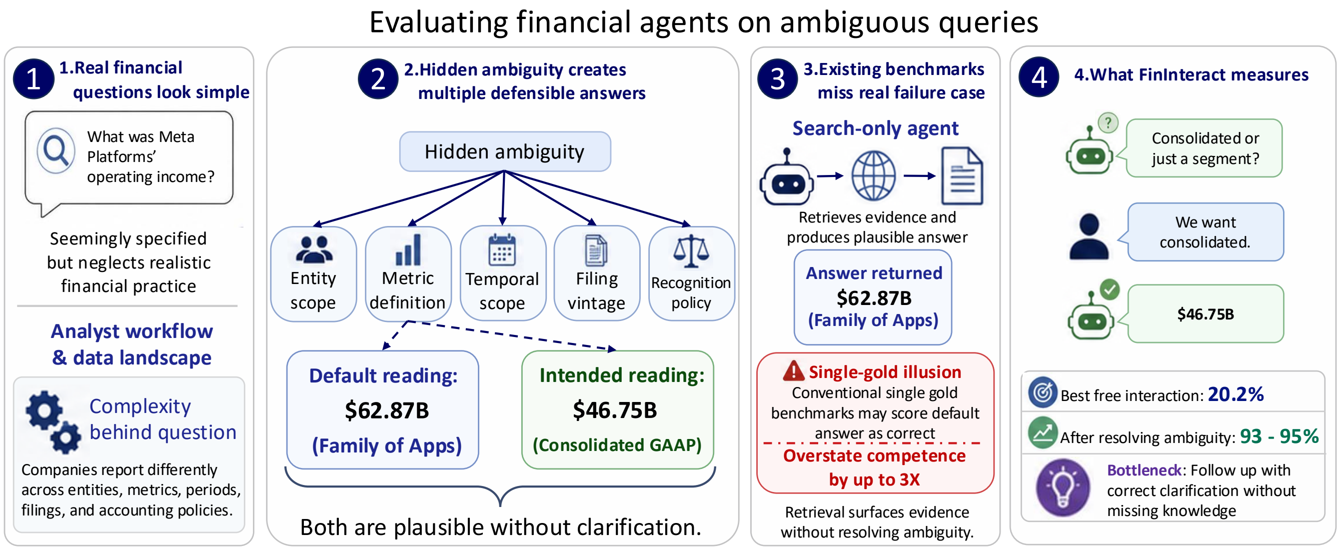

Figure 1: Why ambiguous financial queries need interactive evaluation. (1) A real analyst query ("What was Meta Platforms' operating income?") looks fully specified. (2) But it hides ambiguity along five axes (entity scope, metric definition, temporal scope, filing vintage, recognition policy): here *entity scope* yields two defensible answers—\$62.87B (Family of Apps segment) vs. \$46.75B (consolidated GAAP). (3) A search-only agent retrieves evidence and commits to the plausible default; a conventional single-gold benchmark scores that as correct—the *single-gold illusion*—overstating competence by up to 3×. (4) FinInteract instead measures whether an agent resolves the ambiguity by asking: the best free-interaction accuracy is 20.2% while the context-oracle ceiling is 93–95%, so the bottleneck is issuing the right clarification, not missing knowledge.

Table 1: Running example: a FinInteract instance. The same question yields two defensible answers. Without the disambiguating context C, a model's default interpretation leads to the wrong answer.

Field	Value
Q	What was Meta Platforms' operating income?
C	Total company (consolidated) results under U.S. GAAP for the year ended December 31, 2023.
A (intended)	\$46.75B *[consolidated GAAP]*
A_d (default)	\$62.87B *[Family of Apps segment]*
Ambiguity axis	Entity scope
H_0	1.58 bits
Intended span	"Income from operations for 2023 was \$46.75 billion..."
Default span	"Family of Apps...Income from operations \$62,871..."

multi-answer formulations; CondAmbigQA [12] added condition-structured annotations; CLAM [10] demonstrated that clarification pipelines improve accuracy on ambiguous queries. However, all of this work operates on general-domain Wikipedia questions. Finance is structurally different in three ways. First, financial ambiguity is *endemic and systematic*—a single number such as "net income" can legitimately differ by 30–40% depending on reporting scope, accounting basis, and filing vintage. Second, *misanswers are costly*—an investor acting on the wrong revenue figure or a compliance officer citing the wrong filing version can cause real financial harm. Third, *ground truth is programmatically verifiable*—SEC filings and XBRL facts provide exact machine-readable values for every metric, period, and scope.

The Interaction Stagnation Hypothesis. InteractComp [3] established an important longitudinal finding: while retrieval accuracy on web search benchmarks has improved rapidly with model scale, interaction capability has not kept pace. Across eight frontier models on general-domain tasks, fewer than 14% of questions were answered correctly in full-interaction mode, even though forced-interaction experiments showed the same models could achieve 60–70% accuracy when required to elicit disambiguating context. We hypothesize that this stagnation is more severe in finance, where *domain knowledge is required to even recognize the axes of ambiguity*: a model that does not know what "non-GAAP" means cannot recognize that "adjusted net income" is under-specified.

Our Proposal. We operationalize the single-gold illusion and measure its consequences. We release FinInteract, a bilingual (English/Chinese) benchmark of 173 instances (53 English / 120 Chinese): each question Q is paired with a *default* interpretation (the reading a non-expert assumes) and an *intended* interpretation fixed by a disambiguating context C, across a five-axis taxonomy of financial ambiguity. Grading the *same* model outputs against the

default versus the intended interpretation exposes the illusion directly; a per-axis skill metric then localizes *which* clarification ability a model lacks. Finance is the instrument—chosen for exact, verifiable ground truth and endemic ambiguity—but the phenomenon it reveals is a general fact about how we grade ambiguous queries.
Contributions:

(1) **The single-gold illusion.** We identify and quantify a measurement-validity failure: grading ambiguous queries against one gold answer overstates model competence by up to 3×, because it credits the default reading and cannot distinguish recognizing ambiguity from guessing it. We measure it precisely in finance (§6.1) and confirm it replicates on the general-domain AmbigQA benchmark across three models, where it overstates competence 2.1–2.3× (§6.2).
(2) **An elicitation bottleneck, not a knowledge gap.** Given the resolved interpretation, models answer 93–95% correctly, yet the best model resolves only 20% under free interaction, and *adding* clarification turns *lowers* accuracy below plain search for every model but the strongest—isolating elicitation as the binding constraint (§6.4).
(3) **A per-axis skill decomposition and a capability dissociation.** We turn accuracy into a skill metric (AxisHit) that localizes *which* clarification ability fails, and show the ambiguity axes are different *kinds* of skill: some are solved by scale alone while others (recognition basis) resist scale and are only unreliably teachable (§6.6).
(4) **FinInteract, the instrument.** A bilingual benchmark of 173 verifiable, ambiguity-paired instances with a five-axis taxonomy, construction pipeline, and evaluation harness that make the above measurable and reproducible (§4).

2 TASK DEFINITION

2.1 Problem Formulation

Each FinInteract instance is a tuple (Q, C, A, A_d, $\mathcal{I}_{\text{int}}$, $\mathcal{I}_{\text{def}}$):

- Q is an ambiguous financial question over a public company filing.
- C is a disambiguating context: the minimal set of scope or definitional constraints that uniquely collapses Q to A.
- A is the correct answer under the *intended* interpretation $\mathcal{I}_{\text{int}}$.
- A_d is the correct answer under the *default* interpretation $\mathcal{I}_{\text{def}}$ (the answer a non-expert reading Q alone would assume), with $A_d \neq A$.
- $\mathcal{I}_{\text{int}}$, $\mathcal{I}_{\text{def}}$ are structured records specifying entity scope, period, metric, and reporting basis.

An agent reading Q alone (without C) must find it genuinely ambiguous—both A and A_d are defensible from publicly available filings. This *default-vs-intended pairing* generalizes InteractComp's target-distractor design to the financial domain.

2.2 Design Principles

Easy to verify, Ambiguous to resolve. Adopted from InteractComp: ground truth is programmatically verifiable against SEC XBRL facts and CNINFO-structured data, yet the question cannot be correctly resolved without the disambiguating context C.

No yes/no questions. Q must elicit a substantive financial value, not a binary confirmation.

Company must be named in Q. The question is anchored to a specific issuer (English: capitalized proper noun; Chinese: CJK company name).

Context must not contain the answer. C specifies scope constraints but not the numeric value A itself.

2.3 Agent Interaction Model

We follow InteractComp's ReAct framework [20]. The agent may issue three types of actions: **search**(q)—retrieves a passage from the filing corpus; **interact**(q)—presents a yes/no question to a simulated user who answers based on C; **answer**(v)—submits a final answer. We evaluate agents in three primary modes (Table 2) and four ablation modes.

Table 2: Evaluation modes.

Mode	Actions	Description
Answer-only	answer	Pure parametric recall
Answer+Search	search, answer	Oracle retrieval augmented
Answer+Search+Interact	search, interact, answer	Standard full interaction
Axis-Aware ReAct	state, search, interact, answer	**Proposed: axis-conditione**
Always-ask	Must ask ≥ 1 question before answering	
Axis-oracle	Told the primary axis, not the value	
Template-oracle	Fixed human-written question per axis	
Enumerate	Lists all interpretations and answers	

3 FINANCIAL AMBIGUITY TAXONOMY

We define five axes of financial ambiguity, each grounded in standard accounting and disclosure practice (Table 3).

Table 3: Five-axis financial ambiguity taxonomy.

Axis	Definition
Temporal scope	Ambiguity over which time period is intended: fiscal year ending September vs. calendar year; Q3 results vs. full year; TTM vs. point-in-time.
Metric definition	Ambiguity over accounting basis: GAAP net income vs. non-GAAP adjusted income; organic revenue vs. as-reported; operating EBITDA vs. net income.
Entity scope	Ambiguity over consolidation level: segment vs. consolidated total; parent-attributable vs. full consolidated; Class A vs. Class B shares; A-shares vs. H-shares.
Filing vintage	Ambiguity over filing version: original 10-K vs. amended 10-K/A; as-reported vs. restated; preliminary earnings release vs. final filing.
Recognition policy	Ambiguity over accounting policy: revenue recognized point-in-time vs. over time; gross vs. net revenue; capitalized vs. expensed R&D.

Each instance exercises one primary axis (mandatory) and optionally one or two secondary axes where the passage genuinely supports additional ambiguity. Viable structured ambiguity is far more available for some axes than others: cleanly paired, independently verifiable ambiguity is abundant for entity scope and metric definition but rare for the other three in public filings (Appendix A). The released corpus reflects this—entity scope (54%) and metric definition (37%) dominate, with recognition policy (5%), temporal scope (4%), and filing vintage (0%) sparse. We therefore position FinInteract as an **entity/metric-dominant** financial-ambiguity benchmark with exploratory coverage of three sparser axes, confine quantitative per-axis claims to the two well-powered axes, and report the sparser axes with their sample sizes as preliminary signal.

Disambiguation Entropy. For each instance we compute the prior disambiguation entropy: $H_0 = \log_2 k$, where k is the number of plausible interpretations before any clarifying interaction. For a single-axis instance with two interpretations, $H_0 = 1$ bit. For a two-axis instance with three and four plausible values respectively, $H_0 = \log_2(3\times4) \approx 3.58$ bits. This quantity anchors our supplementary efficiency metric DisE$^+$ (§5.3) and serves as a per-instance difficulty proxy.

4 DATASET CONSTRUCTION

FinInteract is built and evaluated in four stages (Figure 2): we **(1) construct** ambiguous instances from public filings, **(2) verify** their quality with sanity checks, an adversarial verifier, and expert review, **(3) evaluate** agents under a ReAct interaction protocol with a user simulator that answers only from the disambiguating context C, and **(4) score and diagnose** with accuracy plus interaction-quality, calibration, and efficiency metrics. This section covers stages 1–2; §5 and §6 cover stages 3–4.

4.1 Data Sources

English (EN, ~80% of pool). *DocFinQA*—780 long-context passages extracted from 10-K and 10-Q filings, with verified candidate answers from annotated QA pairs. *EDGAR (XBRL-derived)*—111 passages built from machine-readable XBRL facts for S&P 500 and Russell 1000 companies (FY2024–2025). Because instances are newly constructed from structured facts requiring fresh ambiguity pairs, memorization risk is substantially reduced relative to benchmarks that reuse published QA pairs directly—though we do not claim absolute contamination safety.

Chinese (ZH, ~20% of pool). *CNINFO/akshare*—237 passages derived from A-share annual reports (上证 50 + 沪深 300 constituents) via structured financial statement APIs. Three passage types: (1) metric definition—扣非净利润 vs. 归母净利润 (non-recurring-excluded profit vs. parent-attributable profit); (2) entity scope—归母净利润 vs. 净利润 (parent-attributable vs. fully consolidated including minority interest); (3) temporal scope—year-over-year revenue growth. ZH instances are expected to be harder for current models due to sparser training data on A-share filings and more complex corporate structure nomenclature (e.g., A/H-share distinctions, 集团 vs. 子公司 scope).

4.2 Three-Role Construction Pipeline

Instance construction follows a three-role pipeline. Each passage attempt costs approximately 10–14 minutes of wall time, dominated by two sequential GPT-5 calls.

Step 1: Constructor (GPT-5, ~2–5 min). A single GPT-5 call with a 4,096-token budget generates the full (Q, C, A) triple given a filing passage, verified candidate answer, and mandatory primary ambiguity axis. The output JSON includes `question`, `context`, `answer`, `default_answer`, `intended_evidence_span`, `default_evidence_span`, `intended_interpretation`, `default_interpretation`, and `axes_exercised`. Axis-specific instructions are injected per primary axis to prevent the constructor from drifting toward whichever axis is easiest to generate.

Step 2: Pre-verifier sanity checks (~instant). Thirteen automated code-level rules are applied without any API calls (Table 4). Any failure immediately rejects the passage, avoiding wasted verifier budget.

Step 3: Adversarial Verifier (10 trials, parallelized, ~5–10 min). Ten verifier calls fire simultaneously via a thread pool: five GPT-5-mini trials and five GPT-5 trials. Each trial asks the model to answer Q *without* context C and to state the interpretation it assumed (period, entity, metric, basis). An instance is **rejected** if ≥ 2 of the 10 trials both (a) produce the correct answer A and (b) state an assumed interpretation that aligns with the intended interpretation. This two-condition criterion prevents false rejections from lucky guesses: if a model produces the right number while assuming a different interpretation (e.g., the default), the coincidence is not counted against the instance. Because all ten calls run concurrently, wall time equals the latency of the slowest single GPT-5 call.

Table 4: Pre-verifier sanity checks (14 rules).

Rule	Description
R1	Q must not be answerable yes/no
R2	Q must not contain disambiguating terms or inline dates
R3	A taken verbatim from the source candidate value (not invented)
R4	Q must name the company
R5	Q must ask about a substantive financial metric
R6	Answer type consistent with question type
R7	Context C must not contain the answer value
R8	Intended and default interpretations must differ
R9	All axis labels must be valid vocabulary items
R10	Answer must be non-trivial (not empty or "0")
R11	`default_answer` present and differs from `answer`
R12	Both evidence spans non-empty
R13	Evidence spans must be meaningfully distinct (≤85% overlap)
R14	Intended and default *answers* differ beyond grader tolerance

4.3 Axis Diversity Enforcement

A naive construction loop over-represents axes common in the passage pool. We enforce target axis shares through four mechanisms: (1) a priority function with a 3× penalty on over-quota axes; (2) hard exclusion at 2× quota; (3) rarity-adjusted initial sort (passages

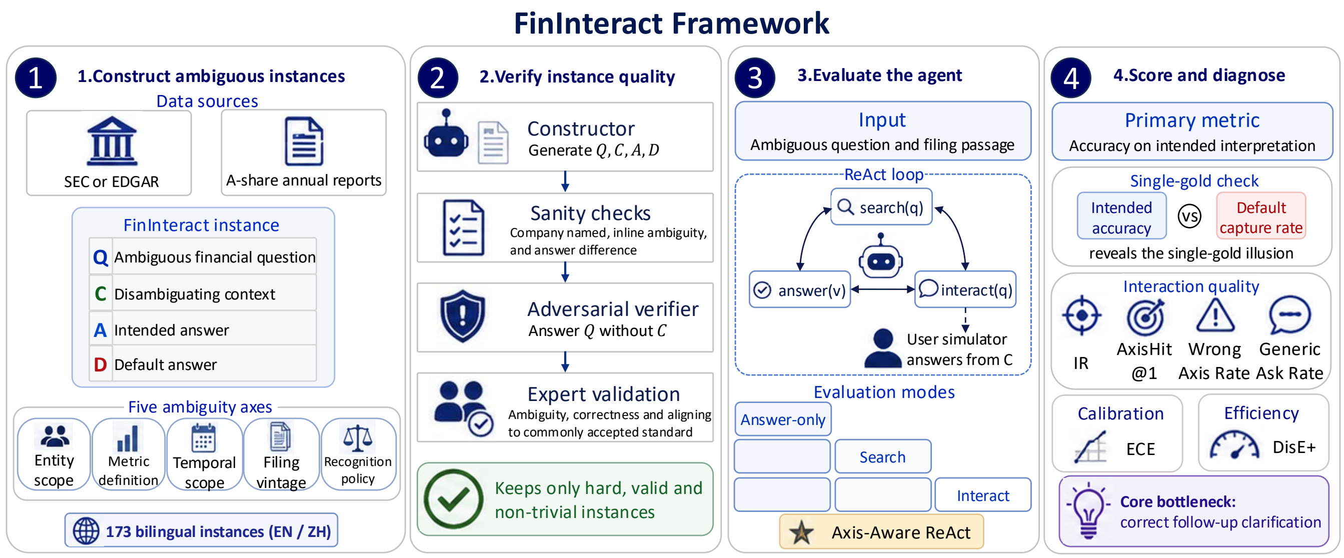

Figure 2: The FinInteract framework. (1) Construct: from SEC/EDGAR and A-share annual reports we build instances, each pairing an ambiguous question Q, disambiguating context C, intended answer A, and default answer D, across five financial ambiguity axes (173 bilingual instances). (2) Verify: a constructor generates (Q, C, A, D); sanity checks, an adversarial verifier (must *fail* to answer Q without C), and expert validation retain only hard, valid, non-trivial instances. (3) Evaluate: agents run a ReAct loop (search/interact/answer) where a user simulator answers only from C, under answer-only, +search, +interact, and Axis-Aware ReAct modes. (4) Score and diagnose: accuracy on the intended interpretation is primary; the intended-vs-default comparison exposes the single-gold illusion, and interaction-quality (IR, AxisHit@1, wrong-axis / generic-ask rates), calibration (ECE), and efficiency (DisE$^+$) localize the core bottleneck—issuing the correct follow-up clarification.

sorted by $\sum_{\text{axes}}(\text{target\_share}/\text{pool\_frequency})$); and (4) dynamic re-sort every 10 instances.

4.4 Human Validation Protocol

Two finance-literate annotators (one bilingual, for the Chinese split) independently review instances on an eight-question protocol (H1–H8): H1 *Is Q ambiguous without C?* H2 *Is the default interpretation plausible for a non-expert?* H3 *Does C uniquely identify the intended interpretation?* H4 *Does C avoid directly stating the answer?* H5 *Is A correct under the intended interpretation?* H6 *Is A_d correct under the default interpretation?* H7 *Is the primary axis label correct?* H8 *What yes/no question would a financial analyst naturally ask to resolve the ambiguity?* (free text, used to compute human AxisHit). Acceptance requires H1–H6 to hold after adjudication by the primary author.

Agreement. On a stratified 60-instance sample (by language and axis), per-item raw agreement is 0.82–0.97. Because the labels are heavily skewed toward "yes" ($>85\%$ per item), Cohen's κ is degenerate under the Feinstein–Cicchetti prevalence paradox; we therefore report the prevalence-robust Gwet's AC1, which is 0.85 **on average** (range 0.78–0.97) and 1.00 for the axis label (H7)—substantial-to-near-perfect agreement. Only one instance was rejected by *both* annotators on the same item (consensus rejection 1.7%); adjudication confirmed it as an axis over-reach (a revenue-recognition-timing pairing whose plausible default is the firm's total revenue, not a disaggregation component). We quarantine it from the public release (`v1.1`, N=172); all experiments below are computed on the frozen `v1.0` evaluation snapshot (N=173), which retains it. The instance is unsolved by every model in every *interactive* mode—it is answerable only when the resolving context is supplied outright (context-oracle), where removing it shifts the ceiling by <0.05pp (93–95% unchanged)—so no reported finding depends on it.

4.5 Dataset Statistics

The frozen evaluation snapshot (`v1.0`) contains 173 instances (Table 5; the public `v1.1` release is N=172, §4.4), drawn from 61 unique companies and SEC/CSRC filings dated 2022–2025. Every instance passes all 14 pre-verifier rules and the 10-trial adversarial verifier (mean blind-solve rate 0.7%), and is R14-discriminating (intended $\neq$ default answer beyond grader tolerance).

5 EVALUATION FRAMEWORK

5.1 Agent Architecture

We evaluate agents under the ReAct framework in seven configurations (Table 2). The **interact** action presents a yes/no question to a simulated user. The user simulator (GPT-5 at temperature 1.0) is given the disambiguating context C and responds yes, no, or "I don't know." To test robustness, we evaluate under three simulator variants: **LLM** (primary; GPT-5, $T = 1.0$), **Oracle** (deterministic keyword matching against $\mathcal{I}_{\text{int}}$), and **Noisy** (15% random answer corruption).

Table 5: FinInteract dataset statistics. Disambiguation entropy $H_0 = \log_2 k$ over k plausible interpretations.

Property	Value
Total instances	173
Language (EN / ZH)	53 / 120
Source (EDGAR / CSRC-CNINFO / DocFinQA)	50 / 120 / 3
Filing type (10-K / 年报)	53 / 120
Unique companies	61
Filing years (2022/23/24/25)	40 / 53 / 58 / 19
Primary axis: entity_scope	94
metric_definition	63
recognition_policy	9
temporal_scope	7
filing_vintage	0
# axes (single / two)	128 / 45
H_0 (mean / median / range)	2.20 / 2.00 / 1.0–3.6
Mean blind-solve rate (verifier)	0.7%

5.2 Models

Eight models are evaluated across all three primary modes: GPT-5, GPT-5-mini, GPT-4o (OpenAI); Claude-Sonnet-4, Claude-Opus-4 (Anthropic); DeepSeek-V3.1, Qwen3-235B, GLM-4.5 (open-weight).

5.3 Metrics

We deliberately lead with metrics that are standard in the ambiguous-QA and interactive-retrieval literature (accuracy, calibration, interaction counts), so that FinInteract is directly comparable to prior benchmarks rather than self-referential. A single bespoke efficiency score is reported only as a supplementary diagnostic.

Accuracy (%) (primary) is the fraction of instances whose final answer matches the *intended* interpretation, graded by GPT-4o-mini with financial-domain tolerance (±1% numeric, entity/ticker equivalence, currency normalization; fiscal-year mismatch = wrong). This is the metric on which models are ranked, mirroring InteractComp, BrowseComp, and FinanceBench.

Default-capture rate (%) is the fraction whose final answer instead matches the *default* interpretation. It measures how often a model confidently returns the naive reading, and lets us reconstruct what a conventional single-gold benchmark would have scored (§6.1).

Interaction diagnostics (secondary, descriptive): *IR* (interaction rate, % of instances with $\geq$ 1 ask), *Round* (mean turns), and *AxisHit*—for each interact action GPT-4o-mini classifies whether the clarifying question targets a true ambiguity axis for that instance; we report AxisHit@1, AnyAxisHit, WrongAxisRate, and GenericAskRate. These characterize *how* a model interacts; they do not rank models.

Calibration: 5-bin Expected Calibration Error (ECE) between stated confidence and correctness, capturing overconfidence on ambiguous queries.

DisE$^+$ (supplementary). As a single number folding accuracy and clarification cost together we report the efficiency of *successful* clarification,

$$\mathrm{DisE}^{+} = \mathbf{1}[\mathrm{correct}] \times \mathbf{1}[n_{\mathrm{asks}}>0] \times \frac{H_0}{n_{\mathrm{asks}}},$$

which is positive only when the agent asked at least one question and then answered correctly, and decreases with redundant asks. We restrict it to the asked-and-correct case precisely to avoid the perverse incentive of an all-instances variant, which would award its maximum to a confident *zero-ask* answer—rewarding the very behavior the benchmark penalizes. DisE$^+$ is a diagnostic of clarification efficiency, never the ranking metric.

6 EXPERIMENTS

6.1 Why Existing Benchmarks Miss This: The Single-Gold Illusion

Before analyzing model behavior, we establish the core motivation empirically: *a conventional single-gold financial-QA benchmark cannot see the failure FinInteract measures.* Existing benchmarks (FinanceBench, FinQA, DocFinQA) attach one gold answer per question. Built from the same filings, such a benchmark would almost always adopt the *default* reading—the headline figure a non-expert assumes—as that gold. Our paired construction lets us quantify the consequence directly: every instance carries both an intended and a default answer, so we re-grade the *identical* model outputs under both conventions (Table 7).

Finding 0: A single-gold convention over-states competence by up to 3×, and the gap is the benchmark's reason to exist. In the retrieval-augmented regime where current single-gold benchmarks operate, the larger models return the *default* reading more often than the intended one: GPT-4o is graded 37.0% correct against the default but only 12.1% against the intended answer—a single-gold benchmark would report 3.1× its true competence. The model is not reasoning to the right answer; it is confidently producing a well-grounded answer to the *wrong reading of the question*, which a single-gold benchmark rewards and therefore cannot detect. (GPT-5-mini is the instructive exception: it captures *neither* reading because it asks rather than commits (Finding 6), so single-gold grading does not flatter it—but nor does it answer.) Critically, interaction is the *only* mode where intended accuracy exceeds default-capture (GPT-5: 20.2 vs. 12.1): asking is what converts a default-anchored guess into the correct intended answer. This is precisely the capability single-gold benchmarks are structurally blind to, and what FinInteract is built to measure.

6.2 The Illusion Is Not Finance-Specific: A General-Domain Replication

Finding 0b: the single-gold illusion is domain-general. Running the identical default-vs-intended measurement on AmbigQA [14] (Table 8) reproduces all three effects outside finance, *consistently across three models.* Grading the *same* answer-only outputs against any accepted reading rather than a specified target overstates competence by 2.1–2.3× (gpt-4o-mini 2.3, gpt-4o 2.2, gpt-5-mini 2.1), and the effect is robust to how the target reading is chosen—a random-target assignment gives 2.1×, versus our most-specific-reading default. Models return a valid but non-target reading 55–66% of the time—direct evidence the error is ambiguity-blindness, not noise.

Table 6: Main results. Accuracy (%) graded against the intended interpretation is the primary metric (per mode); the interact mode additionally reports the secondary interaction diagnostics IR (% asked), AxisHit@1, and ECE. DisE^{+} and the default-capture comparison are reported separately (Table 7). AxisHit is undefined when no question is asked. The OpenAI tier ladder and two open MoE models (Qwen3-30B-A3B, Qwen3.5-35B-A3B) are evaluated; remaining other-vendor rows in progress. The human baseline is the single interaction condition (passage + one yes/no clarification + answer) on a stratified 50-instance subset; see §6.5. Open-MoE +Search/+Interact accuracy is measured over the full retrieval corpus; for both MoE models—as for GPT-4o—added interaction *underperforms* plain search, while only GPT-5 converts interaction into a gain.

	Acc (%)		Answer+Search+Interact			
Model	Ans-only	+Search	Acc	IR	AHit@1	ECE
GPT-5	0.6	6.4	**20.2**	98.8	.38	.53
GPT-4o	0.6	12.1	4.6	99.4	.37	.82
GPT-5-mini	0.6	0.0	0.0	96.0	.34	.00
Claude-Sonnet-4	–	–	–	–	–	–
Claude-Opus-4	–	–	–	–	–	–
DeepSeek-V3.1	–	–	–	–	–	–
Qwen3-235B	–	–	–	–	–	–
GLM-4.5	–	–	–	–	–	–
Qwen3-30B-A3B	1.2	28.3	11.6	72	.16	–
Qwen3.5-35B-A3B	0.6	**35.8**	28.9	36	.40	–
Proposed method baseline (GPT-5 backbone)						
Axis-Aware ReAct	–	–	–	–	–	–
Human (50-inst)	–	–	**30.0**	100	**.68**	–

Table 7: The single-gold illusion. The same model outputs, graded against the *intended* interpretation (FinInteract) vs. the *default* reading that a conventional single-gold benchmark would treat as gold. "Δ" is the inflation a single-gold benchmark would report. In the retrieval regime where prior benchmarks operate, models capture the default far more than the intended answer; only interaction reverses this.

Model	Mode	Intended (true Acc)	Default (single-gold)	Δ
GPT-5	Answer-only	0.6	2.3	+1.7
	Answer+Search	6.4	10.4	+4.0
	+Interact	**20.2**	12.1	−8.1
GPT-4o	Answer-only	0.6	0.6	0.0
	Answer+Search	12.1	**37.0**	**+24.9**
	+Interact	4.6	6.4	+1.8
GPT-5-mini	Answer-only	0.6	0.0	−0.6
	Answer+Search	0.0	0.0	0.0
	+Interact	0.0	0.0	0.0

Table 8: The single-gold illusion replicates on AmbigQA (general domain, n=200, three models, *same* grading methodology; the grader retries with backoff so no answer is silently dropped). *Intended*: accuracy against a specified target reading. *Any-valid*: accuracy against *any* accepted reading—what a lenient single-gold benchmark credits. *Def.-cap*: fraction producing a valid but non-target reading. *Ask*: fraction that asks when permitted. gpt-5-mini interact omitted (it asks <1%).

Model	Mode	Intended	Any-valid	Def.-cap	Ask
gpt-4o-mini	Answer-only	31.5	72.5	55.0	0
	Oracle	49.5	68.5	31.0	0
	Interact	43.5	81.0	53.0	27.0
gpt-4o	Answer-only	39.0	84.0	65.5	0
	Oracle	68.5	81.5	31.5	0
	Interact	43.0	86.0	64.0	2.5
gpt-5-mini	Answer-only	36.0	77.0	62.5	0
	Oracle	62.0	73.5	27.5	0

Elicitation is again underused: even permitted to ask, models clarify on at most 27% of ambiguous queries and usually far less (gpt-4o 2.5%, gpt-5-mini <1%), and interaction leaves intended accuracy far below the any-valid ceiling (gpt-4o-mini 43.5 vs. 81). The one honest difference is the oracle ceiling—50–69% here versus 93–95% in finance—because open-domain factoids stay hard even once disambiguated, so general knowledge is itself a limiter. This is exactly why finance is the sharper instrument: its exact, injectable ground truth isolates *elicitation* from *knowledge* in a way general-domain QA cannot. The illusion itself, however, is a property of single-gold grading, not of finance.

6.3 Findings

We report results for the full OpenAI tier ladder—GPT-5, GPT-4o, and GPT-5-mini—across all three modes (Table 6); other-vendor models are in progress. All headline accuracy comparisons below are reported with bootstrap 95% CIs over instances, and we test each within-model mode transition with a two-sided paired bootstrap (n=173). Six findings emerge here, with a seventh (the latent-capacity ceiling) in §6.4.

Finding 1: Interaction is a capability, not a free lever — it helps the strong model and *harms* the weak one. Adding interaction raises GPT-5 from 6.4% to 20.2% (95% CI [14.5, 26.6]; paired bootstrap p<0.001), but *lowers* GPT-4o from 12.1% to 4.6% (p=0.002). Forcing the weaker model to interact makes it significantly worse than not interacting at all. This extends InteractComp's "interaction stagnation" into a sharper claim: for models that cannot use it, interaction is actively harmful.

Finding 2: Model rankings flip with the interaction axis. GPT-4o *outperforms* GPT-5 under pure retrieval (12.1 vs. 6.4) yet is *dominated* by it under interaction (4.6 vs. 20.2). A single-mode benchmark would mis-rank these models; the interaction axis is what separates them, which is precisely the dimension FinInteract is designed to measure.

Finding 3: Retrieval cannot substitute for interaction. Oracle retrieval lifts GPT-5 to only 6.4%: because each disambiguating passage contains *both* the intended and default values, retrieval surfaces the ambiguity but does not resolve it. Only a targeted clarification collapses the question to a unique answer, which interaction (20.2%) begins to do.

Finding 4: The weak model's failure is clarification non-integration, not inability to ask. GPT-4o asks readily (IR 99%, 4.7 questions on average) but fails to *integrate* the user's responses: 87% of its interactive errors are non-committal refusals (e.g. "please specify the exact period") issued *after* the simulator has answered. It can recognize ambiguity and ask, but cannot close the loop and commit — a multi-turn integration gap, not a protocol artifact (only 2% produced no answer at all).

Finding 5: Models are severely overconfident on ambiguous queries. Expected Calibration Error ranges from 0.53 (GPT-5, interact) to 0.95 (GPT-4o, answer-only); models report 85–95% confidence while answering 0–20% correctly. Both models also ask imprecisely — AxisHit@1 is only 0.38 (GPT-5) and 0.37 (GPT-4o), so even when a model asks, it targets the correct ambiguity axis barely a third of the time, motivating the axis-conditioned policy of §7.

Finding 6: A smaller model collapses into ill-formed clarification — it asks the most but commits the least. GPT-5-mini scores 0.6/0.0/0.0% across the three modes. The cause is not ambiguity-blindness but the opposite: it is the *most* eager to clarify (interact IR 96%, 4.6 asks/instance), yet it issues *open-ended* questions ("which fiscal year? please paste the 2023 and 2022 figures") rather than the protocol's yes/no questions. Manually inspecting all 173 interact trajectories, 63% terminate on an unanswered open-ended question and 35% exhaust the 10-round budget still asking; only 1 instance ever commits a value (also wrong), and just 1.3% were format-unparseable. Even in Answer+Search—where no interact channel exists—it asks instead of answering, capturing *neither* the intended nor the default reading (0.0/0.0%). This is an extreme form of Finding 4's integration gap and a distinct failure mode: clarification without commitment.

Language and axis breakdown. Splitting by language confirms the benchmark's bilingual-difficulty hypothesis: Chinese instances are consistently harder than English in the interact mode (GPT-5 18.3% ZH vs. 24.5% EN; GPT-4o 0.8% vs. 13.2%), reflecting more complex corporate-structure naming and metric conventions in A-share filings. Stratifying by primary axis (interact mode), `metric_definition` is the hardest well-populated axis (GPT-5 12.7%, n=63) and `entity_scope` the next (23.4%, n=94), while `temporal_scope` is comparatively easy (57.1%, n=7). This inverts the general-domain pattern where entity disambiguation dominates, and identifies metric-definition ambiguity as the distinctive financial challenge.

Difficulty and headroom. Per-instance accuracy falls monotonically with disambiguation entropy H_0 (from 10% at H_0=1 to under 1% at $H_0\approx$3.6), validating H_0 as a difficulty proxy; **125 of 173 instances (72%) are solved by no model in any mode**. The best configuration reaches 20.2%, leaving the benchmark far from saturated. The supplementary clarification-efficiency metric mirrors the accuracy ranking in interact mode (DisE^{+} = 0.125 for GPT-5 vs. 0.039 for GPT-4o): GPT-5 not only answers more ambiguous queries correctly but does so with fewer wasted asks.

Table 9: The disambiguation bottleneck. Forcing more clarification does not help (and often hurts), but *resolving* the ambiguity lifts every model to 93–95%. The ≈75-point gap between the oracle ceiling and free interaction is attributable to the model's inability to elicit C itself—not to missing knowledge or grounding. (†GPT-5 forced capped at n=65. Open-MoE interact accuracy is now measured over the full retrieval corpus.)

Model	Answer-only	Interact	Forced(4)	**Oracle ceiling**
GPT-5	0.6	20.2	1.5†	**95.4**
GPT-4o	0.6	4.6	0.0	**93.1**
GPT-5-mini	0.6	0.0	0.6	**95.4**
Qwen3-30B-A3B	1.2	11.6	–	**90.2**
Qwen3.5-35B-A3B	0.6	28.9	–	**91.9**

6.4 Latent Capacity: Forced Interaction vs. the Oracle Ceiling

To locate the bottleneck we bracket interactive accuracy with two ablations (Table 9). **Forced interaction** requires the agent to ask n=4 clarifying questions before answering. **Context-oracle** hands the model the disambiguating context C (which fixes the intended interpretation) together with a passage containing *both* the intended and the default evidence spans, then asks it to answer—the ambiguity is pre-resolved but the model must still ground the correct value.

Finding 7: Models can answer once the ambiguity is resolved—they cannot resolve it themselves. The context-oracle ceiling is 93–95% for all three models, including GPT-5-mini, which scores 0% under free interaction. Latent answering capacity is therefore near-ceiling and roughly scale-invariant; what varies—and what FinInteract measures—is whether a model can *elicit* the disambiguating context. Crucially, *forcing* clarification does not recover this capacity: requiring four asks leaves GPT-4o and GPT-5-mini near 0% (0.0% and 0.6%) and *lowers* GPT-5 from 20.2% to 1.5%, because forced asks consume the round budget on low-value questions without converging on a committed answer. The bottleneck is targeted, self-initiated disambiguation, which neither scale nor brute-force asking supplies.

6.5 Human Baseline

To bound the task from above under *realistic* interaction—where the disambiguating context must be *elicited*, not handed over—we collect a human baseline on a stratified 50-instance subset (32 EN / 18 ZH, balanced by axis). Two finance-literate annotators (one bilingual) face the same input a model does in the search regime: the ambiguous question and a retrieved passage containing both candidate values, position-shuffled and unlabeled. Each may ask *one* yes/no question, which the experimenter answers truthfully from the withheld context C (Yes/No/IDK), before committing a final answer. This mirrors the agent's `search`→`interact`→`answer` loop, so the human row is directly comparable to the +Interact column; we grade it with the identical GPT-4o-mini grader.

Finding 8: Even humans who can ask resolve only 30%**—the task is hard, not merely mis-specified for models.** Annotators reach 30.0% accuracy overall (34.4% EN, 22.2% ZH), above every model's ≤ 20.2% but far from the 93–95% context-oracle ceiling. The gap between the oracle ceiling and the human baseline is the cost of *elicitation*: knowing the resolved answer is near-trivial, but extracting the right bit through a single clarification is hard even for experts. Humans ask far more on-target questions than any model (AxisHit 0.68 vs. 0.34–0.38) yet still capture the default reading 50% of the time. The asymmetry is sharpest in Chinese: ZH annotators ask the *best*-targeted questions (0.72 AxisHit) but post the *lowest* accuracy (22.2%) and highest default-capture (72.2%). Chinese instances are therefore harder to *resolve*, not harder to *recognize*—the 归母/扣非 (parent-attributable vs. deduction-of-nonrecurring) net-income distinction is sticky even once correctly surfaced—corroborating the language-asymmetry caveat in §9 with a mechanism.

6.6 Per-Axis Skills: Which Clarification Abilities Do Models Lack?

The five axes are not just data strata; each names a distinct *skill*—the ability to recognize that a question is under-specified *along that axis* and ask the question that resolves it. We make this measurable by decomposing each model's per-axis `+interact` competence into three components: **Recognition** (interaction rate—does it ask when it should?), **Targeting** (AxisHit@1—does it ask about the *right* axis?), and **Resolution** (accuracy—does asking yield the intended answer?). A model "has" an axis skill only when all three are high; a low value localizes the failure to not-asking, asking-wrong, or asking-right-but-still-wrong. Table 10 reports Targeting, the decisive component.

Finding 9: Ambiguity clarification is a per-axis skill, and models lack specific ones. Targeting is sharply differentiated: every OpenAI model scores AxisHit = 0 on `recognition_policy` (they essentially never ask whether a figure is recognized point-in-time vs. over time), yet all reach ≥ 0.95 on `temporal_scope`; entity and metric sit in between (0.13–0.41). Resolution follows the

Table 10: Per-axis Targeting skill (AxisHit@1) in `+interact`. Recognition policy is a near-universal blind spot; temporal scope is largely solved; entity/metric are partial. "–": model never asked on that axis (IR= 0). The two sparse axes (n≤9) are preliminary.

Axis	n	GPT-5	GPT-4o	GPT-5-m	Q3-30B	Q3.5-35B
entity_scope	94	.41	.29	.37	.28	.39
metric_definition	63	.32	.13	.27	.18	.36
recognition_policy	9	**.00**	**.00**	**.00**	**.00**	.17
temporal_scope	7	.95	1.0	.95	–	1.0

Table 11: Zero-shot Targeting (AxisHit@1) by base-model scale (Qwen3, `+interact`; denominator = instances where the model asked). Entity and metric scale steeply—entity is nearly solved at 32B—while recognition policy stays 0 at *every* scale, including the same 32B that targets entity well. The rare axis is underpowered (n≤9 asked-on) but flat throughout.

Axis	4B	14B	32B
entity_scope	.12	.30	**.89**
metric_definition	.05	.10	.45
recognition_policy	**.00**	**.00**	**.00**

same shape (e.g. GPT-5 resolves 57% of temporal but ≤ 23% of entity/metric). The gap is therefore not uniform incompetence but *missing specific skills*—which is precisely the target an intervention should aim at (§7.4). We also tested whether failures cluster by axis in per-instance behavior space (PCA over per-model correct/ask/AxisHit/default); they do not (k-means ARI ≈ 0.01, p ≈ 0.15), a floor effect of the near-zero accuracies—the axis structure lives in the aggregate targeting profile, not in coarse per-instance behavior. Clustering the *text of the asked clarifying questions* is the sharper test and is left to future work.

Finding 11: the axes are different *kinds* of skill—some scale-solvable, one a scale-invariant blind spot. Zero-shot targeting across a controlled scale ladder (Qwen3 4B → 32B; Table 11) separates the axes sharply. `entity_scope` and `metric_definition` rise steeply with scale—entity is nearly *solved* at 32B (0.89)—so there capability is largely a matter of scale. `recognition_policy`, by contrast, is 0.00 at every scale, including the same 32B model that has clearly learned to target entity: more parameters buy no recognition-basis targeting at all. The dissociation confirms the taxonomy carves genuine capability boundaries rather than arbitrary strata, and it pinpoints where scale stops helping—the axis a targeted intervention must address (§7.5).

6.7 Mechanistic Case Study: Where Ambiguity Lives in the Network

The behavioral results (§6) show that frontier models default to the naive interpretation rather than clarifying. A natural question is whether models internally *represent* this ambiguity at all. Because this requires access to hidden states, we probe four open models as a proxy, spanning two size classes and two architectures: **Qwen3-4B-Instruct** (dense, 37 layers), **Qwen3.5-4B** (hybrid linear/full attention, 33 layers), and the mixture-of-experts **Qwen3-30B-A3B** (49 layers) and **Qwen3.5-35B-A3B** (41 layers).

Method. For each instance we form a within-item contrast: the bare ambiguous question Q versus Q concatenated with its disambiguating context C. We define a per-layer *ambiguity direction* as the difference in mean final-token activations between the two conditions, and score any hidden state by its (mean-centered) projection onto that direction; tracing this across layers gives a depth profile of how strongly each layer separates an under-specified question from a disambiguated one. Independently, on the two larger models we train a per-layer linear probe to decode the ambiguity *type* (entity_scope vs. metric_definition), which—unlike a context-presence detector—cannot be solved trivially.

Result. Figure 3(a) overlays the depth profiles (normalized for differing depth and scale). On all four models the signal is near-zero through most of the stack and rises sharply to a peak in the *final few layers* (≈95–98% relative depth: layers 35/37, 32/33, 48/49, 40/41). Figure 3(b) shows that the ambiguity *type* is linearly decodable at 0.80 accuracy (majority baseline 0.60) on both larger models, though it becomes available at different depths—early (≈27%) in Qwen3-30B and late (≈85%) in Qwen3.5-35B. Across four models, two scales, and three architecture families, the representation is consistent: financial-query ambiguity, and its axis, is linearly encoded, emerging toward the top of the network.

Representation, elicitation, and the cost of interacting. The two MoE models we probe are also *evaluated* (Table 6): beyond linearly encoding the axis, both *elicit*, and they differ in *how* they ask. Qwen3-30B interacts on 72% of instances but loops without committing in Chinese (76% of its interactive transcripts never produce a final answer), whereas Qwen3.5-35B interacts less (36%) but commits in 95% of cases and targets the gold axis more often (AxisHit 0.40 vs. 0.16). With the full retrieval corpus in place their accuracy is now measured, and it exposes a sharper pattern: plain search reaches 28.3%/35.8%, but *adding* interaction *lowers* accuracy to 11.6%/28.9% (−16.8/−6.9 points). Interaction is thus net-negative relative to search for both open models—as it is for GPT-4o (12.1 → 4.6)—and only the strongest model, GPT-5, converts clarification into a gain (6.4 → 20.2). An interpretation-oracle control (the resolved interpretation supplied but no evidence) scores ≈ 0% for both MoE models, so the residual gap to the oracle ceiling is an evidence-grounding cost, not a parametric-recall one. The representational and behavioural evidence shows the axis is both registered and acted on at the level of asking. A linear-probe replication on the *base* (non-instruct) checkpoints reproduces the decodability and the early-vs-late depth split (peaks 0.71/0.74 vs. baseline 0.60).

Finding 10: Open models linearly represent both ambiguity and its axis, replicating from 4B to 35B and across dense and MoE architectures. The representation is type-specific (entity vs. metric decodable at 0.80) and robust to scale and architecture, yet behaviorally these same models still default rather than clarify. The behavioral gap is thus better read as a failure to *act* on a represented ambiguity than a failure to register it—though establishing this causally (e.g. via activation steering) remains future work.

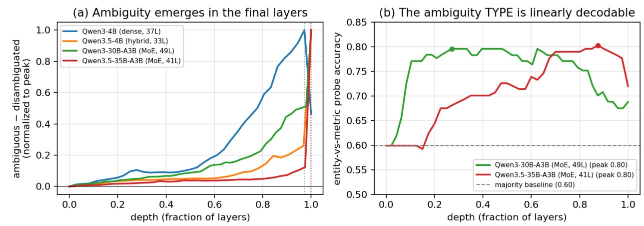

Figure 3: Four-model depth profile of the ambiguity signal (Qwen3 4B→35B, dense and MoE). (a) Ambiguous−disambiguated separation rises to a peak in the final layers of every model (dotted verticals mark per-model peaks; normalized for depth and magnitude). (b) The ambiguity *type* (entity vs. metric) is linearly decodable at 0.80 on both MoE models, well above the 0.60 majority baseline, peaking at different depths.

Scope and caveats. This is mechanistic evidence on open proxies, not the closed frontier models benchmarked in §6; diff-in-means probing and linear decoding establish *availability*, not causal use; and axis-type decoding is supported only for the two well-populated axes (entity_scope, metric_definition). We deliberately do *not* draw behavioral conclusions from these open models' free-form generations, where surface cues (chain-of-thought markers, retrieval that surfaces the evidence span) confound naive clarification counts; the asking-behavior claims come solely from the graded closed-model evaluation in §6. That the axis is *represented but not acted on* frames the bottleneck as an action problem on an available signal—and motivates the axis-conditioned interventions of §7, which target exactly that signal.

7 DIAGNOSIS AND INTERVENTION: ACTING ON THE AMBIGUITY AXIS

Our results converge on a single diagnosis. The benchmark localizes the bottleneck to *eliciting* the disambiguation: agents answer 93–95% correctly once the intended interpretation is supplied (§6.4) but resolve only ∼8% on their own, and the mechanistic probe shows why—open models *linearly represent* the ambiguity *axis* yet do not act on it (§6.7). The remaining gap is thus an *action* problem on an already-available signal, and the *axis* is its natural unit. This section turns that diagnosis into intervention: we first catalog how acting fails (a seven-type error taxonomy), then present two interventions that share one idea—**condition on the axis**—instantiated at inference time (Axis-Aware ReAct, §7.3) and at training time (axis-guided SFT/GRPO, §7.4).

7.1 Error Taxonomy

Beyond aggregate accuracy, we characterize model failures using a seven-type error taxonomy (Table 12) derived from FinInteract's evaluation signals. Each type is diagnosable from existing metrics without additional annotation.

Table 12: Error taxonomy E1–E7 with diagnostic signals and improvement directions.

Type	Description	Improvement direction
E1	**Ambiguity blindness**: no interaction, wrong answer	Ambiguity pre-check; train on detection
E2	**Wrong-axis clarification**: asked wrong dimension	Axis classifier; financial taxonomy grounding
E3	**Generic clarification**: vague ask, no axis target	Axis-conditioned clarification templates
E4	**Retrieval dominance**: searched but skipped interaction	Force interpretation check after retrieval
E5	**Clarification misuse**: right question, wrong answer	State-gated answering; stronger grounding
E6	**Evidence grounding**: template-oracle correct, agent wrong	Improved source-grounded extraction
E7	**Over-interaction**: correct but low DisE^{+}	Penalize via DisE^{+}/AxisHit efficiency signal

7.2 Diagnostic Mapping from Evaluation Signals

The existing evaluation metrics map directly onto error types, enabling failure attribution without additional annotation:

- **Low IR + low Acc** → E1 (ambiguity blindness dominant)
- **High IR + low AxisHit@1** → E2/E3 (interacts but misdirected)
- **High AxisHit@1 + low Acc** → E5/E6 (right axis, wrong grounding)
- **High GenericAskRate** → E3 (lacks financial clarification structure)
- **High WrongAxisRate** → E2 (confidently misdirected)
- **Always-ask** ≫ **Standard ReAct** → E1 dominant; latent capacity exists
- **Template-oracle** ≫ **Agent-interact** → "what to ask" is the bottleneck
- **Axis-oracle** ≫ **Agent-interact** → axis recognition is the bottleneck
- **Template-oracle still low** → E6 residual; evidence extraction still fails

7.3 Inference-Time Intervention: Axis-Aware Clarification ReAct

The first intervention conditions on the axis *without changing the model weights.* Standard ReAct leaves the agent free to decide whether and what to ask. We propose **Axis-Aware Clarification ReAct**, a three-component structured interaction policy that directly addresses E1–E3.

Component 1: Ambiguity Pre-Check. Before any action, the agent explicitly decides which of the five ambiguity axes are underspecified in the question and initializes an interpretation state $\mathcal{S}$ = {entity, period, metric, basis, vintage}, marking each field as *resolved* or *unknown*.

Component 2: Axis-Conditioned Clarification. For each *unknown* field, the agent generates a targeted yes/no question using axis-conditioned templates (one per axis) rather than free-form clarification. The agent emits at most one clarification per turn, prioritizing the most information-theoretically valuable unknown field.

Component 3: Interpretation-State-Gated Answering. The agent may only emit a final answer once all required fields are resolved (or the user explicitly declines to clarify). This prevents the *retrieval dominance* pattern (E4) where the agent answers immediately after finding a plausible number in a search result.

Expected outcomes. Relative to standard Answer+Search+Interact, Axis-Aware ReAct should reduce GenericAskRate (E3), WrongAxisRate (E2), and IR=0 failures (E1), while improving AxisHit@1 and ultimately Accuracy. Template-oracle provides an upper bound: if Axis-Aware ReAct approaches template-oracle performance, it confirms that axis recognition is the dominant remaining bottleneck after the E1–E3 errors are addressed.

7.4 Training-Time Intervention: Axis-Guided SFT and GRPO

The per-axis skill analysis (§6.6) gives this intervention a precise target: models do not lack clarification ability uniformly—they lack *specific* axis skills (AxisHit = 0 on recognition policy, partial on metric). The question is whether those missing skills can be installed by training rather than prompting. The second intervention applies the *same* idea—condition on the axis—but in the weights rather than the prompt, showing the benchmark is not only diagnosable but *trainable*. Starting from Qwen3-4B-Instruct (4-bit QLoRA), we run an RLVR ladder—supervised fine-tuning (SFT), KTO, then GRPO—on the FinInteract training split, with all user simulators, graders, and axis judges served locally (zero API cost). We score each stage with the two *leak-proof* metrics, AxisHit@1 and interaction rate (Table 13); accuracy is reported only with the caveat that the training environment's retrieval returns the disambiguated evidence span, so it is partly leaked and is *not* a reliable learning signal.

The axis-guided teacher. The decisive ingredient is how SFT demonstrations are generated. An uninformed teacher—even a 72B model—almost always asks the *obvious* ambiguity (typically the reporting period) and rarely the subtle gold axis (entity scope, recognition policy), yielding ~0% on-axis demonstrations. Privately revealing the gold axis to the teacher while generating demonstrations (the hint is *not* stored in the trajectory) lifts the on-axis yield to ~79%—a reusable recipe for any axis-structured clarification task.

SFT does the heavy lifting. The leak-proof signals jump almost entirely at the supervised stage—AxisHit@1 rises 25 → 69 and interaction 55 → 100 (Figure 4a)—while KTO and GRPO move within run-to-run noise at n=51. Training teaches the model to *always interact and target the correct axis*, the behavior FinInteract is built to elicit. Whether this gain reflects a sharpened internal *representation* of the axis or only a rewired output *policy*—probing the trained model as in §6.7—is a direct test of the diagnosis and a target for future work.

True multi-turn GRPO. The ladder above optimizes a single-turn approximation. We additionally confirm the full multi-turn

Table 13: RLVR ladder on FinInteract (Qwen3-4B, held-out n=51). AxisHit@1 and interaction are leak-proof signals of learned clarification; accuracy (*) is partly leaked by oracle retrieval and should not be read as a learning signal.

Stage	AxisHit@1	Interaction	Accuracy*	Reward
Base (Qwen3-4B)	25.0	54.9	84.3	0.96
+ SFT (axis-guided)	68.6	**100**	88.2	1.28
+ KTO	**70.6**	**100**	84.3	1.27
+ GRPO	66.7	**100**	88.2	**1.30**

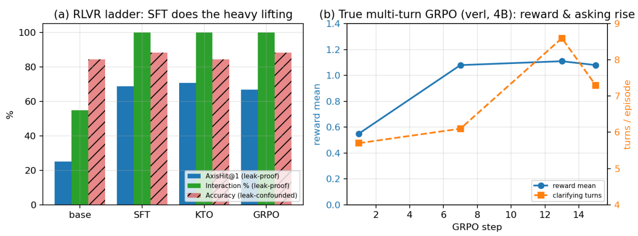

Figure 4: Training on FinInteract. (a) The leak-proof learning signals (AxisHit@1, interaction) jump at SFT and saturate; accuracy (hatched) is leak-confounded. (b) True multi-turn GRPO on the 4B policy: mean reward and clarifying turns rise together over training.

episode is trainable end-to-end: optimizing the 4B policy over entire ReAct episodes (loss masked on simulator and retrieval tokens) for 15 steps raises mean reward 0.55 → 1.08 while clarifying turns rise 5.7 → 8.6 (Figure 4b)—the axis-aware reward is teaching the policy to ask more. We report this as a learning-signal demonstration, not a converged model.

Caveats. Results are on a held-out split of n=51; all judges and the teacher are local Qwen2.5-32B/72B models (weaker than GPT-4o, and siblings of the policy); the terminal reward is partly leaked; and GRPO is run for tens of steps, not to convergence. We therefore frame this as evidence that FinInteract is *trainable*—an axis-guided SFT seed instills axis-targeted clarification—rather than as a state-of-the-art accuracy result.

7.5 When Can the Benchmark Refresh Itself? Three Co-Evolution Probes

A benchmark that models eventually memorize stops measuring. Because FinInteract couples a grounded data-generation pipeline (§4) to a trainable policy (§7.4), it can in principle *co-evolve*: generate fresh instances on an axis the current model fails, train on them, and repeat—a data↔model loop in the spirit of self-improving generators [2], but *grounded* in real filings rather than the ungrounded self-play those methods risk. We run three gated rounds—on the scarce, subordinate axis (`recognition_policy`), the abundant, salient axis (`entity_scope`), and the abundant-but-only-weakly-scale-learnable axis (`metric_definition`, §6.6)—to ask not *whether* the loop works but *when*.

All three rounds run end-to-end: the Constructor→Verifier pipeline yields hard instances (every one passes the adversarial gate—the

Table 14: Three co-evolution rounds (axis-guided SFT on a 4B policy). Targeting (AxisHit@1) is on a *held-out human* probe never seen in training. The loop closes on both *abundant* axes and fails only on the *scarce* one; whether asking *helps accuracy* flips on whether the base could already answer.

	`entity` abundant	`metric` abundant	`recognition` scarce
Frontier instances generable	99	76	36 (ceiling)
Targeting, base → evolved	.00 → **.68**	.00 → **.58**	.00 → .00
on held-out *generated*	.40	.88	.00
Interaction rate	40 → 100	50 → 100	67 → 100
De-leaked accuracy	32.5 → 22.5	15 → **60**	11 → 11

base fails it answer-only) and axis-guided SFT converges. The outcome splits cleanly by *abundance*, not by scale-learnability. On both abundant axes—`entity_scope` and `metric_definition`—Targeting generalizes to held-out *human* items (.00 → .68 and .00 → .58, each ~4× the +.15 bar) and to held-out *generated* items, with no catastrophic forgetting. On the scarce `recognition_policy` it does not move (.00 on both held-out sets): the trained model reverts to the salient period/entity question rather than the recognition-basis distinction. The contrast rules out a broken pipeline—the *same* machinery installs entity and metric but not recognition—and, because metric is only *weakly* scale-learnable yet trains as well as entity, it rules out scale-learnability as the gate (§7.6).

Finding 12: the loop closes on abundant axes and fails on scarce ones; whether elicitation *helps* is a separate, per-axis property. A self-refreshing FinInteract is viable wherever the data arm can supply enough frontier instances—both abundant axes install and generalize; the scarce axis (recognition's 36-instance ceiling from just 30/1,847 primary passages) does not reliably (Finding 14). But installing on-axis elicitation is not automatically beneficial: it *helps* metric (accuracy 15 → 60, +45—the base cannot answer without the metric definition, so the clarifying question *supplies* it: 18 wrong→correct, 0 regressions) and *hurts* entity (accuracy 32.5 → 22.5, −10—the base can often answer the entity directly, so forcing a question turns six correct direct answers into wrong multi-turn ones). Both drove interaction to 100%; the accuracy sign flips on whether the base could already answer—the training-time face of the "interaction can hurt versus plain answering" effect seen at inference (§6.4). The actionable step is a *conditional when-to-ask gate*—elicit only where the model cannot already answer—plus multi-round curriculum and fresh-filing ingestion to lift the scarce axes above their availability ceiling.

7.6 What Gates Co-Evolvability? A Falsification Test

The three rounds let us isolate *which* axis property predicts whether the loop closes. Two candidates are natural—an axis's *scale-learnability* (does zero-shot Targeting rise with model size?) and its *source-abundance* (can the data arm supply frontier instances?)—and we had also hypothesized a representational version: *learnable iff the axis is linearly decodable* from hidden states. The `metric_definition`

Table 15: What predicts co-evolvability. The observed outcomes track *abundance/salience*—*not* the AxisHit scale-slope (metric: low slope, GO) and *not* decodability (recognition: highest AUROC, yet does not co-evolve). Salience = primary/any-candidate passages; decodability = peak one-vs-rest AUROC (recognition on a size-matched augmented sample; raw n=9 overfits to 1.0). *metric is a GO but needs ~2 accumulated slices (~50 instances); one 30-instance slice teaches it nothing (Finding 14). ‡recognition is *unreliable* rather than a clean fail: held-out human AxisHit ranges 0–0.44 across training runs on the scarce n=9 probe.

Axis	Primary	Salience	Slope	Decode	Co-evolve
entity_scope	905	**0.92**	+0.77	0.77	**GO**
metric_definition	498	0.48	+0.21	0.77	**GO***
temporal_scope	150	0.15	sat.	0.92	solved
filing_vintage	264	0.33	–	–	untestable
recognition_policy	30	**0.04**	+0.00	**0.93**	weak‡

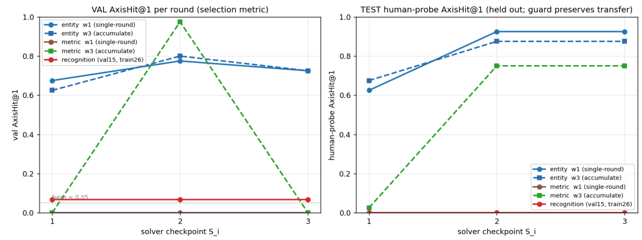

Figure 5: Multi-round co-evolution with a validation guard. *Left*: validation AxisHit@1 (selection metric) per round. *Right*: held-out *human*-probe AxisHit (never used for selection). Entity learns targeting from a single 30-instance slice and transfers to humans (0.63 → 0.93); metric needs *accumulation* (one slice teaches nothing—brown, flat—while two accumulated slices reach 0.75—green); recognition stays at base (≈ 0.05) even with a fair 26-instance set. The guard preserves human transfer (it rolls back the round-3 candidate, which regresses on validation).

round was designed to discriminate the first two; a linear probe tests the third. Both tidy hypotheses fail.

Scale-learnability does not gate co-evolution. `metric_definition` is abundant (498 primary passages) but only *weakly* scale-learnable (zero-shot AxisHit .05 → .26, slope +0.21, versus entity’s +0.77). We pre-registered the two-factor prediction that it would co-evolve only *partially*. It did not: metric is a *full* GO (Targeting +0.55, matching entity’s +0.61; Table 14). What separates GO from NO-GO is not the scale-slope but *abundance*—both abundant axes close the loop; only the scarce axis fails.

Representation does not gate it either, and is not the bottleneck. A per-layer one-vs-rest linear probe decodes every axis from base hidden states, and the pattern *anti*-tracks learnability: `recognition_policy`—the un-learnable, NO-GO axis—is the *most* decodable (AUROC 0.93 on a size-matched sample, above entity/metric at ∼ 0.77); decodability is *flat* across the 4B → 32B sweep even as behavioural AxisHit climbs .12 → .89; and axis-guided SFT leaves it unchanged. The axis is fully *represented* at every scale, before and after training—the model simply does not *act* on it, the “represents-but-doesn’t-act” gap of §6.7, now shown to hold even for the axis co-evolution cannot fix.

Finding 13: co-evolvability is gated by source *abundance*—driven by *salience*—not by scale-learnability or representational decodability. What survives both falsifications is a single upstream cause. `recognition_policy` is a candidate axis in 796 passages but *primary* in only 30 (salience 0.04): it is almost always the *subordinate* ambiguity, behind a more salient entity or period axis in the same passage. Low salience produces both the scarcity that starves the data arm and the salient competitor the model’s prior reaches for instead—so the representation is present but never acted on and cannot be cheaply trained in. Salient axes behave oppositely (entity, salience 0.92: abundant and trainable). For a practitioner, **salience—not scale curves or probe accuracy—is the cheap, up-front signal** of which axes a co-evolving benchmark can keep hard and which need fresh human curation. The corollary for evaluation is sharper: the hardest ambiguity to make models *handle* is not the one they cannot represent, but the one that is always someone else’s more-obvious question.

Finding 14: salience predicts an axis’s *data-need*, not just a binary outcome—and a validation-guarded arms race confirms it. Extending each axis to a multi-round curriculum (Fig. 5), with checkpoints selected on a frozen validation set and reported on the held-out human probe, refines the binary gate into a gradient that tracks salience. **Entity** (salience 0.92) learns axis-targeting from a *single* 30-instance slice and transfers to human ambiguity at 0.88–0.93. **Metric** (0.48) needs *accumulation*: one slice yields 0.0, two accumulated slices reach 0.75—so its earlier single-round GO used ~60 instances precisely because that is roughly what the axis requires. **Recognition** (0.04) is *unreliable*: across training runs its human AxisHit ranges only 0–0.44 on the scarce n=9 probe—weak and inconsistent, not a clean zero. Scoring the same runs as an *Elo arms race* between the solver and a hard-example miner (win = AxisHit, which—unlike recall-walled resolution—varies across checkpoints) makes the instrument non-degenerate: entity is *solver-runaway* (solver Elo 1000 → 1551, outpacing a corpus-bounded miner that cannot synthesize adversarially harder targeting), while recognition *stalls* (flat). Two guarded rounds peak; the third overfits and is rolled back. The practical reading: salience is an up-front estimate not just of *whether* an axis co-evolves but of *how much* data it needs—and one well-targeted round, guarded on held-out validation, is close to the sweet spot.

8 RELATED WORK

Financial QA Benchmarks. Table 16 compares FinInteract to major financial QA benchmarks. FinanceBench [8], FinQA [5], DocFinQA [16], BizBench [9], FinBen [19], and FinanceQA [13] all enforce single-gold-answer conventions and evaluate models in a single turn; FinanceQA further shows that frontier models fail ~60% of realistic analyst tasks, but attributes this to numerical rigor rather than ambiguity. FinEval [7], a large Chinese financial-knowledge benchmark, incidentally identifies an "ambiguity handling weakness" error type in its agent tasks but does not isolate or control it. Recent *agentic* financial benchmarks raise task realism—Finance Agent Benchmark [4] (537 expert research tasks, where the best model reaches only 46.8%) and FinAgentBench [6] (agentic retrieval over EDGAR)—but both remain single-turn and single-gold, evaluating analysis and retrieval rather than whether an agent recognizes that a query is under-specified. FinInteract is the first to make query ambiguity the central, controlled construct—via paired default/intended interpretations and multi-turn interaction—in the financial domain.

Ambiguous QA.. AmbigQA [14], ASQA [17], CondAmbigQA [12], and CLAM [10] have matured in the general (Wikipedia) domain. More recent benchmarks target the agent's *own* ability to ask: CLAMBER [22] evaluates whether models identify and clarify ambiguous information needs, and QuestBench [11] tests whether they can ask the minimal necessary question in reasoning tasks. Both are general-domain and judge a single clarifying step; FinInteract adds *per-axis* targeting (AxisHit), programmatically verifiable financial grounding, and—via the co-evolution rounds (§7.5)—an analysis of *when* asking helps versus hurts. We adapt this line's formalism to finance, replacing human-judged ambiguity labels with programmatically verifiable XBRL-grounded evidence pairs.

Clarifying Questions. Asking clarifying questions is a long-standing goal in information retrieval and dialogue. Rao and Daumé III [15] rank clarification questions by expected value of perfect information; ClariQ [1] established a shared task for generating clarifying questions in open-domain dialogue; and the mixed-initiative paradigm is surveyed in Conversational Information Seeking [21]. FinInteract instantiates this lineage with a leak-proof, per-axis measure of *which* clarification an agent should issue and a verifiable outcome for whether issuing it resolves the query.

Interactive Search Agents. InteractComp [3] is the direct methodological ancestor. We extend its general-domain setting to finance with a domain-specific ambiguity taxonomy and bilingual evaluation. BrowseComp [18] tracks longitudinal retrieval trends; we test whether the interaction-capability gap identified there persists in finance.

9 LIMITATIONS

Scale. At 173 instances FinInteract is small (comparable to FinanceBench's 150 [8]); we prioritize per-instance construction rigor (every item is adversarially verified and R14-discriminating) over raw size, and Appendix A shows the pipeline scales at $0.31/instance. **Language balance.** The English split (53) is smaller than the Chinese split (120); per-language results (§6) should be read with this asymmetry in mind, and the EN portion is a target for growth. **Axis coverage.** The taxonomy is intentionally five-axis, but `filing_vintage` is unrepresented and `temporal_scope`/`recognition_policy` are sparse (Table 5): viable structured ambiguity for these axes is rare in public filings (Appendix A). **Model coverage.** We report the OpenAI tier ladder (3 models); other-vendor models are in progress. The human baseline (N=50) establishes the performance ceiling but is collected from two annotators on a subset; a larger expert panel remains future work. **Grader and simulator.** Correctness and axis-targeting are judged by GPT-4o-mini and the user is simulated by GPT-5; although 1.3% of interact outputs were format-unparseable and manually checked, a full judge-vs-human agreement study remains future work. **Contamination.** Instances are freshly constructed from FY2022–2025 structured facts and answer-only accuracy is ≈0.6% across all sources, but we do not claim absolute contamination safety for the DocFinQA-derived minority.

10 CONCLUSION

We introduced FinInteract, the first bilingual benchmark for evaluating financial search agents on genuinely ambiguous queries. The five-axis financial ambiguity taxonomy, paired default/intended interpretation design, and adversarial verifier-based construction pipeline yield 173 high-quality instances that resist trivial disambiguation. One unit organizes the whole story—the ambiguity *axis*: it structures the taxonomy, is what an agent must elicit (single-gold grading overstates competence up to 3× when models commit to the wrong axis-reading), is *linearly represented* inside open models yet left unacted-on, and is the lever for intervention. Our central finding is that interaction capability—not financial knowledge—is the bottleneck: agents answer 93–95% correctly once the axis is resolved but ~8% on their own, amplifying the stagnation InteractComp identified in the general domain. We turn this diagnosis into intervention with two axis-conditioned methods—at inference (Axis-Aware ReAct) and at training (axis-guided SFT/GRPO)—and release FinInteract with full construction, evaluation, and training code for reproducible follow-on work.

Table 16: Comparison of financial QA benchmarks. ✓ = supported; – = not applicable or not evaluated.

Benchmark	Scale	Ambiguity	Interaction	Multi-turn	Bilingual	Evidence	Grounding
FinanceBench [8]	150	–	–	–	–	human	SEC
FinQA [5]	8,281	–	–	–	–	human	10-K/Q
DocFinQA [16]	7,437	–	–	–	–	human	10-K/Q
BizBench [9]	multi-task	–	–	–	–	auto	mixed
FinBen [19]	multi-task	partial	–	–	partial	auto	mixed
FinanceQA [13]	multi-task	–	–	–	–	expert	SEC
Finance Agent Bench. [4]	537	–	–	–	–	expert	SEC
FinAgentBench [6]	26K	–	–	–	–	auto	SEC
FinEval [7]	8,351	–	–	–	ZH	expert	exams
FinInteract (ours)	173	✓	✓	✓	EN+ZH	LLM+human	EDGAR+CNINFO

A CURATION COST TO REPLICATE THE BENCHMARK

A practical concern for any LLM-assisted benchmark is the cost to reproduce or scale it. We instrument the construction pipeline (§4) and report measured API costs. All figures use list prices at construction time (per 1M tokens: GPT-5 $1.25 in / $10.00 out; GPT-5-mini $0.25 / $2.00; GPT-4o-mini $0.15 / $0.60) and count API fees only—they exclude the one-time filing-collection compute and the human spot-check labor (10–20% of accepted instances).

The constructor, not the verifier, is the yield bottleneck. Re-running the pipeline over a held-out sample of candidate passages reproduces a sharp funnel (Table 17). Only 11.5% of keyword-tagged candidate passages survive the constructor, which rejects the rest as containing *no viable ambiguity* for the target axis; but of those that survive, 98.5% pass the 10-trial adversarial verifier. In other words, the construction prompt is strict and the verifier rarely needs to fire—the cost of the benchmark is dominated by constructor calls spent on candidate passages that turn out not to support a genuine ambiguity, not by verification.

Table 17: Measured curation cost. Unit costs are from instrumented pipeline runs (constructor-only n=20; full pipeline n=6); the acceptance funnel is from the full construction run (n=1127 candidate passages). A full pipeline pass is one GPT-5 constructor call plus ten verifier trials (five GPT-5, five GPT-5-mini) with GPT-4o-mini grading.

Quantity	Value
Constructor viability (viable / candidate)	11.5%
Verifier acceptance (accepted / viable)	98.5%
Overall acceptance (accepted / candidate)	11.3%
Cost per rejected candidate (constructor only)	$0.020
Cost per viable passage (full pipeline)	$0.155
Blended cost per candidate passage	$0.035
Cost per accepted instance	**$0.31**

Projected cost to scale. At the measured blended rate, producing N accepted instances requires $\approx N/0.113$ candidate passages and the following API spend: 173 instances (the current release) ≈ $54; 500 instances ≈ $155; 1000 instances ≈ $309. The benchmark is therefore inexpensive to scale in API terms—an order of $300 takes it to 1000 instances—and the true constraint is the supply of candidate passages that contain structured, verifiable ambiguity. This favors *better passage pre-selection* (raising the 11.5% viability rate) over brute-force compute, and explains why the rare axes (e.g. filing_vintage), whose candidate passages almost never survive the constructor, are the expensive ones to grow.

B REPRODUCIBILITY

Models and decoding. Closed models are accessed via their official APIs (evaluated 2026): `gpt-5`, `gpt-4o`, `gpt-5-mini`. Reasoning models (gpt-5 family) use default temperature with a generous `max_completion_tokens` headroom (+4096 above the output budget) so internal reasoning does not truncate the answer; other models use temperature 0 for the agent and grader. **User simulator.** GPT-5 at temperature 1.0, constrained to {Yes, No, I don't know} and conditioned on the disambiguating context C. **Grader.** GPT-4o-mini at temperature 0, binary correctness with finance tolerance (±1% numeric, entity/ticker equivalence, currency normalization; fiscal-year mismatch = wrong). **Protocol.** ReAct with `search`/`interact`/`answer` actions, max 10 rounds; the forced-interaction ablation requires n=4 asks before answering; the context-oracle ceiling supplies C plus both evidence spans (position-shuffled). **Statistics.** Accuracy CIs are 95% bootstrap over instances (B=2000); mode transitions use a two-sided paired bootstrap (B=5000). All evaluation, construction, and analysis scripts are released.

C DATASHEET AND LICENSING

Provenance. English instances derive from public SEC EDGAR filings (10-K, FY2022–2025) and a small DocFinQA-sourced set; Chinese instances derive from public CSRC annual reports (年报) retrieved via CNINFO/akshare. All source documents are public regulatory filings; no proprietary, paywalled, or personal data is used. **Schema.** Each instance is a JSON record with fields: `instance_id`, `language`, `source`, `ticker`, `company`, `filing_type`, `filing_date`, `question`, `context` (C), `answer`, `default_answer`, `intended_evidence_span`, `default_evidence_span`, `intended_interpretation`/`default_interpretation` ({entity, period, metric, basis}), `axes`, `n_axes`, `h0`, and `qc` (per-rule pass flags + verifier blind-solve rate). **Intended use.** Evaluating whether interactive search agents recognize and resolve query ambiguity; not a training corpus and not financial advice. **License and release.** The benchmark, construction pipeline, and evaluation code are released for research use; redistribution of source filings follows the respective public-domain (EDGAR) and regulatory-disclosure (CSRC) terms. **Maintenance.** The frozen `v1` release is versioned; corrections and the planned EN-split expansion will be issued as numbered releases.

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